

Design, modeling and optimization of an m-shaped variable reluctance energy harvester for rotating applications

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ABSTRACT

The variable reluctance principle can be used to convert rotational kinetic energy into electrical energy, creating a Variable Reluctance Energy Harvester (VREH) based on electromagnetic induction. This can be used to implement self-sustaining wireless sensors in rotating applications. In this paper, we present and investigate a novel design of a VREH with high volumetric power density that targets low-speed rotating applications. The design uses an m-shaped pole-piece and two opposing magnets. We theoretically analyze key design parameters that influence the VREH's output power, and relate these parameters to geometrical design factors of the proposed structure. Key design factors include the coil height, the permanent magnet height and the tooth height. A method based on numerical simulations is introduced, enabling to determine the optimal geometrical dimensions of the proposed structure under given size-constraints. The results demonstrate that the method leads to optimal structure configurations, which has been evaluated for different cases and is verified experimentally. Good agreement between numerical simulations and experiments are reported with deviations in output power estimation below 3%. The optimized m-shaped VREH, moreover, provides output power levels sufficient for wireless sensor operation, even in low-speed rotating applications.

1. Introduction

Energy harvesting enables the conversion of ambient energy into electrical energy in order to supply power to low-energy electronic devices such as sensor nodes or deeply embedded systems [1,2]. As a result, the usage of purely battery-based solutions can be reduced, leading to beneficial effects on maintenance requirements. Among the available sources, kinetic energy sources have obtained much attention, because of their general availability and relative high energy densities [3,4]. In industrial applications, kinetic energy sources are commonly found in the forms of translational and rotational motion. Although typically providing a more continuous and predictable output, rotational motion has been studied much less than, for example, vibrations.

In order to harvest rotational motion, typically electromagnetic and piezoelectric transduction mechanisms have been utilized. In [5–7], for example, different piezoelectric energy harvesters for rotating applications have been introduced. The presented energy harvesters utilize different forms of cantilever structures that are excited by the alternating direction of the gravitational force relative to the cantilever. While these solutions have the benefit of being easily added to the rotating object, they are sensitive to the rotation frequency, their orientation, as well as other superimposed motions (e.g., shocks or

vibrations). Electromagnetic mechanisms, utilizing Faraday's law of induction, exploit changes in the magnetic flux to induce an electromotive force (EMF) in a coiled conductor [8]. In most existing solutions, this flux change is provided by relative motion between magnets and coils [9].

A variable reluctance energy harvester (VREH) is a type of electromagnetic energy harvester that creates a magnetic flux change by varying the reluctance in a magnetic circuit. As a result, both the magnet and the pickup coil are stationary with respect to a moving ferromagnetic structure, typically implemented as a toothed wheel for rotating hosts. With a constant magnetomotive force (MMF) and a varying magnetic reluctance, an EMF is induced in the pickup coil due to the changing magnetic flux. As a result of its simple structure, variable reluctance energy harvesting is a suitable method in many rotating applications. Because changing shaft dimensions mainly affect the size of the ferromagnetic toothed wheel, it is, moreover, easily scaled to different application scenarios. In contrast, traditional electromagnetic generators require scaling of permanent magnets or coils, making them costly in applications with large rotating shafts and slow rotational speeds.

In an earlier study, we have demonstrated that the structure of the pickup unit in a VREH considerably affects its output power and power

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density [10]. An m-shaped VREH has been introduced, which outperformed all previously reported structures in the literature. The objective of this paper is to provide a detailed study of this novel energy harvesting structure, as well as to provide an optimization method for its parameters with respect to application constraints.

Previous studies have introduced variable reluctance energy harvesting as a means for energy conversion in rotating applications. In [11], the authors experimentally demonstrated that commercial variable reluctance sensors could be utilized as energy harvesting transducers, but did not investigate the transducers optimization for energy harvesting purposes. In [12,13], the authors designed and evaluated a VREH system for railroad applications. In these studies, the effect of coil position on the generated output power of a passing train wheel was tested, but the structure otherwise remained unchanged. A motivation for the utilized configuration was not provided and the energy output of the demonstrated prototype was relatively low.

Based on the survey presented in [10], the selection of an m-shaped VREH leads to the highest power density. It remains, however, the question how different parameters in such an architecture should be chosen in order to generate most power under known application constraints. To answer this question, in this paper the proposed architecture is analyzed in detail, and the effect of different configuration parameters on the output power are investigated based on numerical simulation and experimental verification. In Section 2, a description of the m-shaped structure and its working principle will be provided. In Section 3, we define application constraints as an available volume for the VREH and theoretically analyze the effects of its configuration on output power. The setups of numerical simulations and experiments are described in Sections 4 and 5, followed by the presentation of results in Section 6. Finally, the paper is concluded in Section 7.

2. The m-shaped variable reluctance energy harvester

2.1. Topology description

Fig. 1 illustrates the m-shaped VREH topology. Similarly to other VREH topologies, it comprises two major components, a toothed wheel and a pickup unit. The toothed wheel is a center-hollowed and toothed structure, which is mounted on an existing rotating target such as a rotary shaft. The pickup unit is stationary with respect to the toothed wheel and consists of a pickup coil, two identical permanent magnets (PM), and an m-shaped pole-piece. The pickup coil is wound on the center pole of the pole-piece, and its two terminals are coupled to a load or an energy harvesting power management circuit. An air gap exists between the toothed wheel and pickup unit, which is a necessary mechanical clearance for allowing the rotation of the toothed wheel. The two PMs are placed in close proximity to the air gap at the two outer poles of the pole-piece, facing the air gap with the same pole.

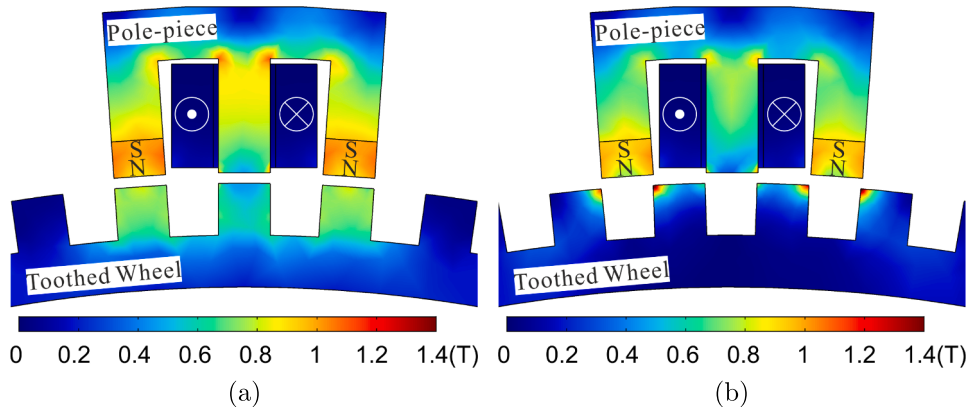


Fig. 1. Proposed VREH topology and the magnetic flux density distribution at its two extreme positions, (a) aligned position and (b) unaligned position.

On rotation, the relative position between toothed wheel and pickup unit changes, which results in periodic arrangements between two extreme positions. These positions are the aligned position, shown in Fig. 1a, and the unaligned position, shown in Fig. 1b. At the aligned position, each pole of the m-shaped pickup unit is fully aligned with a tooth of the toothed wheel, whereas each pole faces a slot of the toothed wheel in unaligned position.

The m-shaped pole-piece and the toothed wheel are made from the same high-permeability, soft ferromagnetic material, constructing a low-reluctance magnetic flux pathway in aligned position. With minimum reluctance in the aligned position, leading to a high flux density in the center pole in Fig. 1a, a strong magnetic coupling of the pickup unit and the toothed wheel with minimal flux leakage is achieved. In contrast, the reluctance is increased to its maximum when the unaligned position is reached, resulting in a lower flux density in the center pole. Utilizing the two magnets in opposite magnetic orientation, the two magnetic flux loops are forced to converge in the center pole, resulting in a high flux level through the surface spun by the pickup coil. With a high flux level, also the flux change is high when changing reluctance in the path.

2.2. Operation principle

Rotating at an angular velocity of ω , the relationship between the mechanical power and electrical power can be expressed as

$$P_m = \omega \cdot \tau = P_{\text{Iron}} + P_{\text{Coil}} + P_L = P_{\text{Iron}} + 2P_L, \quad (1)$$

where P_m is the mechanical power extracted from the rotating body; τ is the torque caused by the electromagnetic force generated by the induced current in the pickup coil; P_{Iron} represents the iron loss; and P_{Coil} and P_L are the power dissipated by the copper coil and the load, respectively. Furthermore, P_{Coil} can be substituted by P_L , in the case of an impedance matched load.

It has been shown in [10] that, following Faraday's law of electromagnetic induction, in a VREH, the electrical power dissipated in the load can be expressed as

$$P_L = \frac{N_{\text{Coil}}^2}{8R_{\text{Coil}}} \Delta\phi^2 \left[\frac{\omega N_{\text{Tooth}}}{2\pi} \right]^2. \quad (2)$$

This indicates that the output power P_L is proportional to the number of coil turns N_{Coil} , the magnetic flux difference $\Delta\phi$ between the two extreme positions, the rotating speed ω , and the number of teeth N_{Tooth} . At the same time it is inversely proportional to the coil resistance R_{Coil} .

Eq. 2 neglects the inductive part of the impedance, which will influence the output power. This is a reasonable assumption for low angular velocities. Higher angular velocities induce higher electrical frequencies of the alternating output voltage, and the impedance match between the pickup coil and load will change with the electrical

frequencies. A complex conjugate impedance matching thus should be considered to achieve the maximum power transfer between the pickup coil and load at higher rotational speed.

Assuming that ω and N_{Tooth} are determined directly or indirectly by the application scenario, two sub-functions of Eq. (2) can be extracted that define the major influences on the output power. These sub-functions are

$$P_L \propto N_{\text{Coil}}^2 / R_{\text{Coil}} \quad (3a)$$

$$P_L \propto \Delta_\phi^2 \Rightarrow P_L \propto (\mathcal{F}_M / \Delta_\phi)^2, \quad (3b)$$

where $\mathcal{F}_M$ and Δ_ϕ are the magnetomotive force (MMF) and the reluctance difference between the aligned and unaligned position, respectively. They are, moreover, related with the flux difference according to

$$\Delta_\phi = \mathcal{F}_M / \Delta_\phi. \quad (4)$$

Furthermore, the MMF $\mathcal{F}_M$ of a magnet can be expressed as

$$\mathcal{F}_M = H_c \cdot h_M, \quad (5)$$

where H_c is the coercivity of the magnet, and h_M is defined as the dimension co-linear to H_c .

Consequently, parameters of the m-shaped VREH that directly or indirectly affect the coil parameters or the flux variation, will affect the output power generation of the configuration.

3. Geometrical influences and structural optimization

Scaling the size of an energy harvesting device is a direct approach to increase the output power, but is typically limited by space and weight constraints in practical applications [4]. Volumetric power density is thus an important metric to indicate the performance of many energy harvesting devices [14]. The goal of the structural optimization in this work is to maximize the output power of the proposed VREH structure for a given volumetric constraint.

3.1. Definition of the volumetric constraint

The volumetric power density p_v of the VREH can be expressed as

$$p_v = \frac{P_L}{\mathcal{V}_{\text{VREH}}}, \quad (6)$$

where P_L is the output power (see Eq. 2), and $\mathcal{V}_{\text{VREH}}$ is the necessary volume for the VREH. $\mathcal{V}_{\text{VREH}}$ can be defined in different manners. As the toothed wheel and the pickup unit are added to the shaft, they will increase the total diameter of the system. Consequently, for the remainder of this work, $\mathcal{V}_{\text{VREH}}$ is defined as a hollow cylinder illustrated in Fig. 2 with

$$\mathcal{V}_{\text{VREH}} = \frac{\pi l_o}{4} (d_o - d_{\text{Shaft}})^2. \quad (7)$$

Herein, d_o and d_{Shaft} are the overall diameter and shaft diameter, respectively, and l_o is the overall depth of the VREH structure.

The added volume $\mathcal{V}_{\text{VREH}}$ can be utilized in different ways, which is influenced by multiple geometrical parameters as defined in Fig. 2. As a simplification, we mainly consider five of these geometrical parameters, namely, the pickup coil height h_{Coil} , the tooth height h_{Tooth} , the slot width W_{Slot} , the air-gap size l_g , and the magnet height h_M . In addition, the following relationships are prescribed for the remainder of this work

$$\begin{aligned} W &= W_y = W_p = W_m = W_t = W_{bk} = W_{cp} \\ A &= W \cdot l_z \\ l_z &= l_{z,M} = l_{z,\text{Wheel}}, \end{aligned} \quad (8)$$

where A is the cross section of the flux path, W is the flux path width, and l_z , $l_{z,M}$, $l_{z,\text{Wheel}}$ are the depth of the pole-piece, magnet and toothed

wheel, respectively. These relations enable the VREH structure to have a uniform cross section for the magnetic flux through the entire path.

The total diameter d_o can be expressed based on d_{Shaft} and the parameters of the VREH, such that

$$d_o = 2(h_{\text{CPole}} + l_g + h_{\text{Tooth}} + W_{bk}) + d_{\text{Shaft}}. \quad (9)$$

Because of the low permeability of air, a larger air gap significantly weakens the variable reluctance effect, causing a reduced output power. The air gap l_g should therefore be set to the minimum clearance that allows the toothed wheel to rotate freely, including considerations on manufacturing tolerances and assembly.

Given a constant l_g and W , as well as a constrained $\mathcal{V}_{\text{VREH}}$, a constant C can be defined as

$$C = h_{\text{Coil}} + h_{\text{Tooth}}, \quad (10)$$

providing an optimization goal in selecting the h_{Coil} to h_{Tooth} ratio that maximizes output power. In order to achieve this structural optimization, an analysis of each height's influence on output power is presented in Sections 3.2 and 3.3, leading to a trade-off further discussed in Section 3.5. Moreover, the magnet height h_M , which does not directly influence $\mathcal{V}_{\text{VREH}}$, will affect the output power of the harvester, as it affects the MMF. Its relationship with output power is analyzed in Sec. 3.4.

3.2. Influences of coil parameters

According to Eq. (3a), output power is proportional to the number of turns and the electrical resistance of the pickup coil. These parameters, in turn, are dependent on the available volume to the coil, the wire diameter, and the utilized winding technology. It can be shown, see Appendix A, that a relationship between geometric coil parameters and output power can be deduced from Eq. (3a), such that

$$P_L \propto h_{\text{Coil}} W_{\text{Coil}} f_w. \quad (11)$$

Herein, W_{Coil} and h_{Coil} are the coil width and height shown in Fig. 2c, and f_w is the winding fill factor.

The product $h_{\text{Coil}} W_{\text{Coil}}$ is the cross sectional area constrained by the pole-piece height h_{CPole} and the pole pitch Δ_{Pole} , which defines the space that can be utilized by the coil. In order to maximize P_L , the coil should fill as much of this area as possible. Consequently, h_{Coil} and W_{Coil} will be

$$W_{\text{Coil}} = \Delta_{\text{Pole}} - (W_{cp} + 2t_{\text{Bob}}) \quad (12)$$

$$h_{\text{Coil}} = h_{\text{CPole}} - (W_y + 2t_{\text{Bob}}), \quad (13)$$

where t_{Bob} is the coil bobbin thickness, necessary to insulate the pickup coil from the pole-piece.

Assuming that W_{Coil} and f_w are fixed values, Eq. (11) can further be simplified to

$$P_L \propto h_{\text{Coil}}, \quad (14)$$

indicating that output power increases linearly with the coil height, and thus the total height of the pickup unit h_{CPole} . As a result, the available coil volume will be increased, improving flux linkage.

3.3. Influences of tooth height

As shown in Eq. (3b), the output power of the VREH is dependent on the flux change between its aligned and unaligned position. Assuming that the MMF of the magnets is constant, this equals to a dependency on the reluctance change. The reluctance of a material is expressed by

$$\mathcal{R} = \frac{l}{\mu_0 \mu_r A}, \quad (15)$$

where μ_0 is the free space permeability, μ_r is the relative permeability of the respective material, A is the cross sectional area, and l is the effective path length. Considering the relative permeability of the

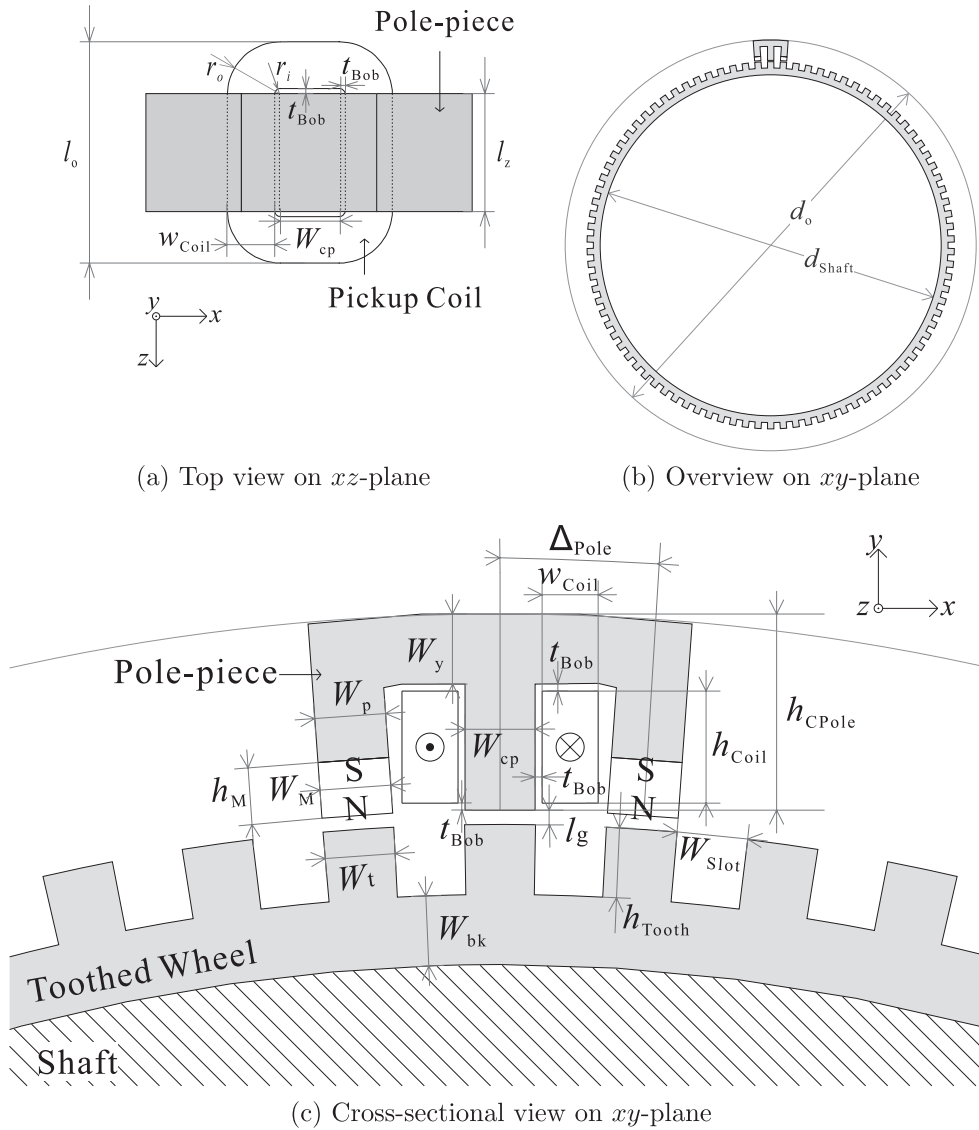


Fig. 2. Definitions of geometric parameters in the proposed topology in (a) top view (only showing the pickup unit), (b) cross-sectional overview of the entire system, and (c) cross-sectional view of the pickup unit.

materials in the flux path of the VREH, the pole-piece and toothed wheel have negligible contributions to the total reluctance due to the large permeability of the soft ferromagnetic material.

Instead, the air gaps ($\mu_r = 1$) and the permanent magnets ($\mu_r = 1.05$) dominate the structures reluctance. This allows to present Eq. (4) in a simplified form, such that

$$\Delta_\phi = \frac{\mathcal{F}_M}{\Delta_\mathcal{R}} \Rightarrow \Delta_\phi = \mathcal{F}_M \left(\frac{1}{\mathcal{R}_{g,a} + \mathcal{R}_M} - \frac{1}{\mathcal{R}_{g,u} + \mathcal{R}_M} \right), \quad (16)$$

where $\mathcal{R}_{g,a}$ and $\mathcal{R}_{g,u}$ are the total air reluctance at the aligned and unaligned position, respectively. With $\mathcal{R}_M$ being constant, Δ_ϕ is thus defined by the variation in air gap reluctance.

At the aligned position, illustrated in Fig. 3, $\mathcal{R}_{g,a}$ of a single magnetic flux loop is the sum of the side-air-gap reluctance and center-air-gap reluctance. With a small l_g , i.e., a strong coupling between magnet and tooth, fringing can be neglected and $\mathcal{R}_{g,a}$ is mainly defined by the shaded areas in Fig. 3[15]. The tooth height practically has no effect.

At the unaligned position, exemplified in Fig. 4, the situation is inverse and most of the magnetic flux fringes to the teeth, whereas basically no flux reaches the bottom of the slot. This, however, is dependent on the tooth height and thus the slot depth. With shorter teeth,

the portion of the flux reaching the bottom of the slot increases. In contrast, longer teeth result in a larger $\mathcal{R}_{g,u}$, increasing the Δ_ϕ and thus output power. Dominated by the first few millimeter of the teeth [16,17], this effect diminishes when continuing to increase tooth height. Results of a numerical simulation of this effect are shown in Fig. 5, validating the previous analysis.

Theoretically, the effect of tooth height on $\mathcal{R}_{g,u}$ and thus output power should saturate, as the flux reaching the bottom of the slot approaches zero. However, Fig. 5 illustrates a continuously increasing output power with almost linear relationship to the tooth height after a certain threshold value. This behavior can be explained based on longer teeth causing a wider slot width W_{slot} at the air gap, when maintaining the same shaft diameter. It has been shown that this increased W_{slot} leads to a large air gap reluctance at the unaligned position [16].

While this has no effect on the reluctance in the aligned position, the reluctance of the unaligned position is increased due to the larger distance between magnets and teeth. Consequently, for the VREH structure, this leads to an increased reluctance difference, maintaining a linear dependency with h_{tooth} after the knee point has been exceeded.

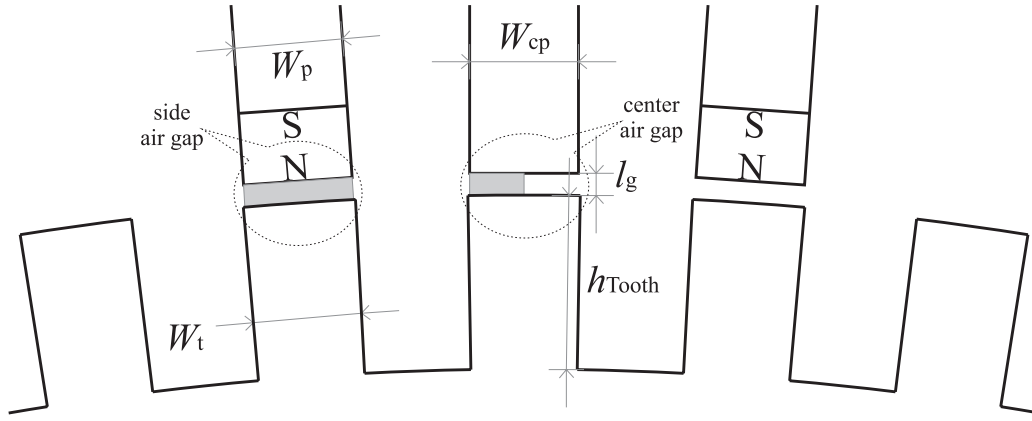


Fig. 3. Idealized air-gap flux distribution of a single magnetic flux loop at the aligned position, which contributes to two major reluctance elements, side-air-gap reluctance, and center-air-gap reluctance.

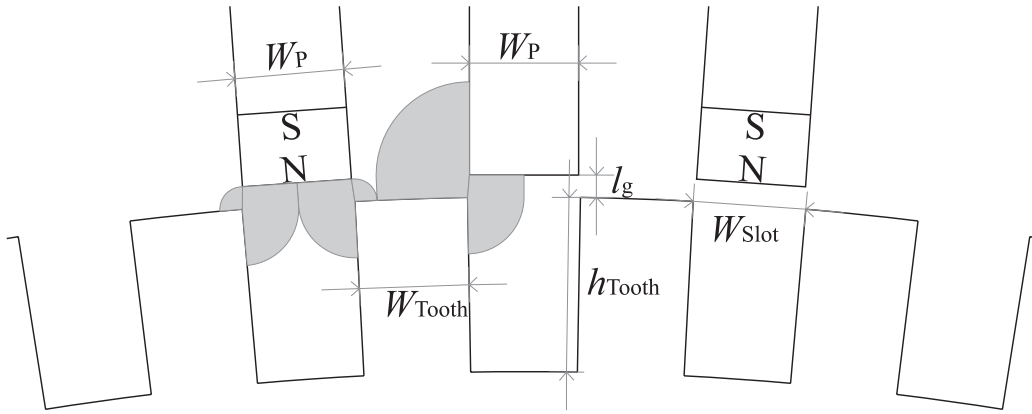


Fig. 4. Idealized air-gap flux distribution of a single magnetic flux loop at the unaligned position. The fringing effect causes no magnetic flux passing through the slots' bottoms.

3.4. Influences of magnet height

Magnet dimensions affect the magnet performance such as the permeance coefficient [18]. As shown in Eq. (5), h_M has a linear effect on the MMF and thus the flux difference, increasing power output. However, h_M also affects $\mathcal{R}_M$ according to Eq. (15).

It can be shown, that an optimal h_M exists, which maximizes $\Delta\phi$, maximizing, in turn, the VREH output power. The optimal h_M depends

on the geometry of the air gap at the two extreme positions and the permeability of the magnet material, and can be derived from Eq. (16) as demonstrated in Appendix B. Accordingly, the optimum magnet height can be calculated as

$$h_M = \mu_M \frac{k_2}{k_1} \sqrt{\frac{W_{g,a}}{W_{g,u}}} l_{g,a} l_{g,u}, \quad (17)$$

where k_1 is the leakage flux coefficient, k_2 is the iron loss factor,

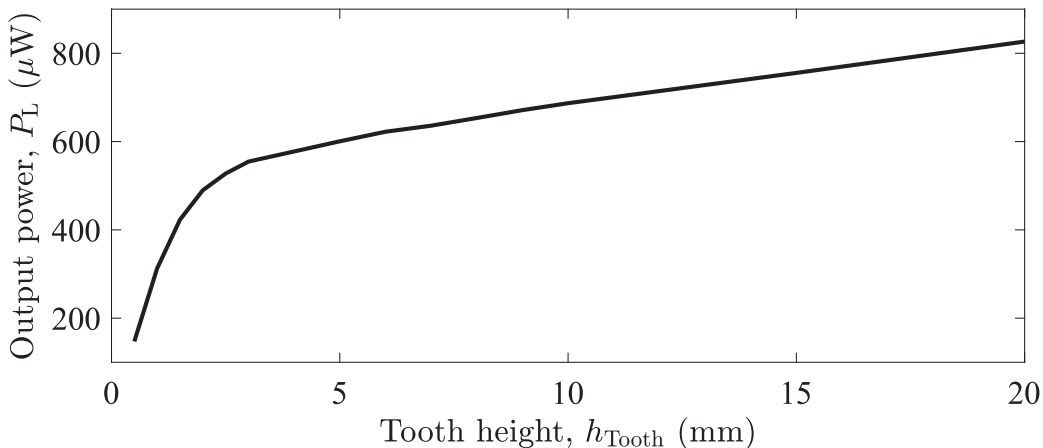


Fig. 5. Numerical sweeps of different tooth heights under constant angular velocity $\omega = 1.06 \text{ rad s}^{-1}$. The output power increases rapidly until a tooth height of approx. 3 mm. Afterwards a slow continuous power improvement is a result of a widened slot width.

$l_{g,a}$, $l_{g,u}$, $W_{g,a}$ and $W_{g,u}$ are the geometrical dimensions of the air gap at the two extreme positions (a : aligned position, u : unaligned position).

3.5. Coil height and tooth height trade-off

As presented in Sections 3.2 and 3.3, h_{Coil} and h_{Tooth} dominate the coil resistance and magnetic reluctance effect, respectively, and both factors thus influence the output power. For a constant C , increasing h_{Coil} improves the output power by expanding the coil cross sectional area that interacts with more magnetic flux, but it has to decrease h_{Tooth} , resulting in a reduced reluctance change. In contrast, increasing h_{Tooth} increases air gap reluctance variation, at a cost of a reduced available area for the pickup coil. For a given γ_{VREH} , an optimal ratio between the two factors thus contribute to an optimized VREH implementation with high volumetric power density.

Because of the leakage flux, the fringing effect, and the non-linearity magnetic performance of soft iron, it is complex to estimate the optimal structure and its output power in a purely analytical manner. Therefore, a numerical study has been conducted and will be presented in Sec. 4. We perform parametric sweeps of the previously discussed factors in order to find the optimal VREH structure. The findings of the numerical simulations are then verified experimentally.

4. Numerical analysis setup

The VREH structure is numerically analyzed based on a 3D electromagnetic model, implemented in the finite element analysis (FEA) software COMSOL Multiphysics, utilizing its AC/DC module. The VREH is a short-axial-length and large-diameter electric machine. Implementing the model in 3D takes into account the end effects, which therefore produces more accurate results [19].

Fig. 6 exemplifies an implemented model with its meshing configuration. Because the model is symmetrical in the z -direction, only half of the total structure (cut along the xy -plane) is simulated, which reduces the model size and the required computational time. All results presented, however, are scaled to the entire structure.

The *Rotating Machinery, Magnetic (rmm)* interface of COMSOL is used to model the rotary motion at a constant angular velocity, $\omega = 1.06 \text{ rad s}^{-1}$. Furthermore, the *Electrical Circuit (cir)* interface obtains the output power by connecting a matched resistive load to the pickup coil.

Temperature effects are neglected in the numerical analysis, and all

Table 1

Geometrical parameters and material specification.

Domain	Parameter	Value
Magnet	h_M	2 mm–8 mm
	W_M	5 mm
	$l_{z,M}$	10 mm
	Material	NdFeB-N50 ($\mu_M = 1.05$, $H_c = 860 \text{ kA m}^{-1}$)
Coil	h_{Coil}	2 mm–14 mm
	N_{Coil}	based on h_{Coil}
	R_{Coil}	based on h_{Coil}
	d_{wire}	0.1447 mm
	W_{Coil}	4 mm
	S_l	0.8443
	S_c	0.7012
	ρ	$16.8 \mu\Omega \cdot \text{mm}$
Pole-piece	W_{cp} , W_p , W_y	5 mm
	h_{cp}	based on h_{Coil}
	l_z	10 mm
	Material	NSC-35H230 [20]
Toothed Wheel	h_{Tooth}	based on h_{Coil}
	W_t	5.06 mm
	$l_{z,\text{Wheel}}$	10 mm
	N_{Tooth}	90
	W_{bk}	5 mm
	Material	NSC-35H230 [20]
Air	l_g	1 mm
	$l_{g,\text{eff}}$	1.2 mm
	t_{Bob}	0.5 mm
	μ_r	1
Shaft	d_{Shaft}	270 mm
	ω	1.06 rad s^{-1}
	Material	Polyoxymethylene (POM) ($\mu_r = 1$)

material properties are modeled to be temperature invariant with material properties for 20 °C.

Table 1 summarizes all domain configurations and modeling parameters, based on realistic material selections and an industrial application scenario. Configurations and parameters are selected in accordance to the experimental setup.

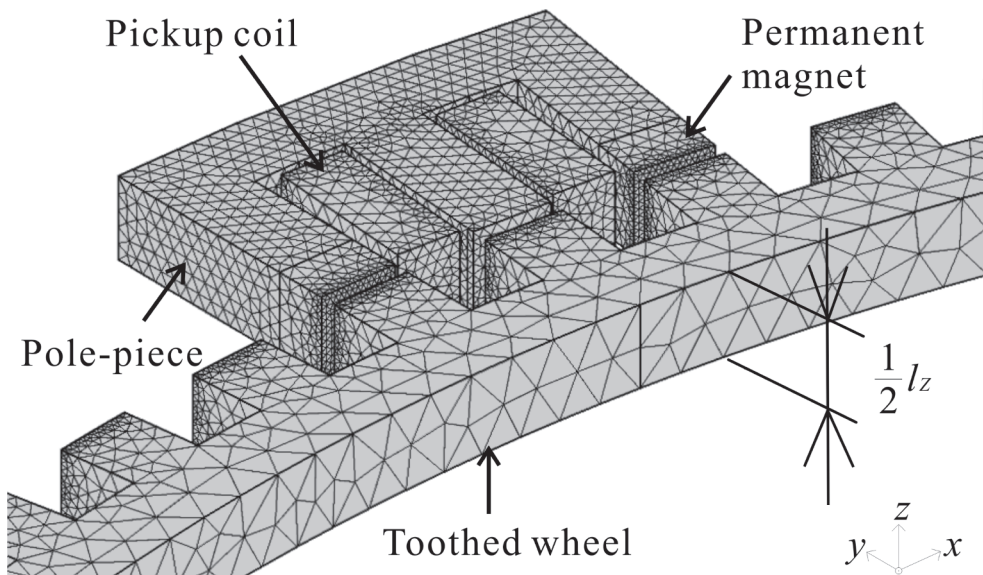


Fig. 6. Exemplar VREH FEA implementation with its meshing configuration. The model is halved along the xy plane due to symmetry conditions.

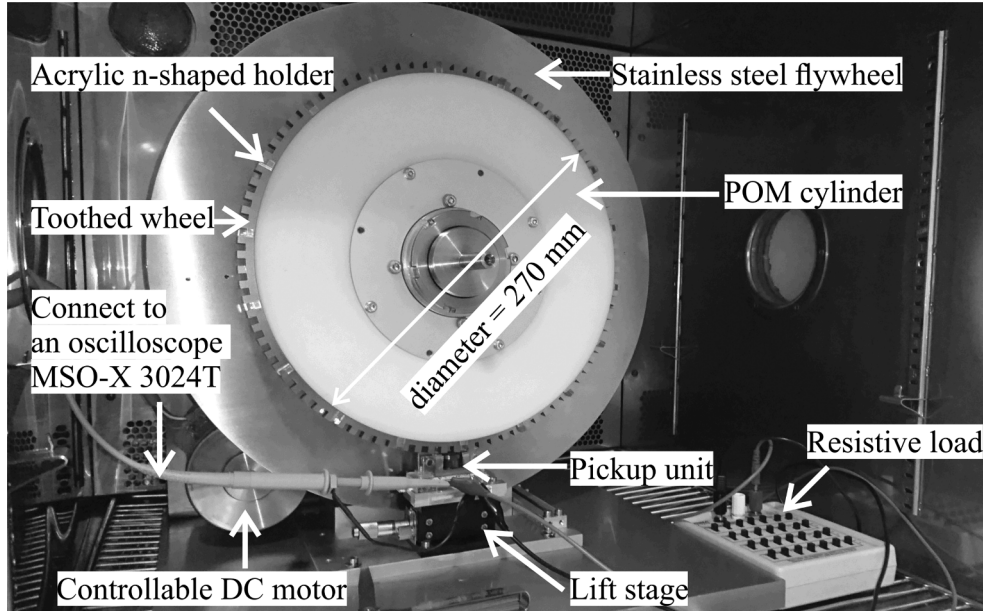


Fig. 7. Experimental setup placed in an environmental chamber (TestEquity 1007H), maintaining a stable ambient temperature of 20 °C.

4.1. Adjustments for fabrication factors

In order to allow for a free motion of the shaft and mechanical assembly while minimizing the reluctance in the aligned position, an air gap length l_g of 1 mm has been selected for simulations and experiments. In the physical implementation of the VREH structure, however, the two rectangular magnets are implemented as a stack of smaller magnets. This fabrication decision will have two effects on the structure.

Firstly, each of the smaller magnets has a coating, which adds length to the non-ferromagnetic path. This coating has an approximate thickness of 0.05 mm. For simplicity, its permeability can be estimated to be that of air. Secondly, the magnets are glued to each other, as well as to the pole-piece, creating additional gaps. These effects can be accounted for by creating an effective air gap length $l_{g,eff}$.

In the presented configuration, the total additional air gap length is estimated to be 0.2 mm, which is taken into account as $l_{g,eff}$, listed in Table 1.

4.2. Adjustments for iron losses

The output power computed by the FEA does not consider iron losses. Consequently, the provided value is not the true output power, and we denote it as P'_L . Due to impedance matching $P'_L = P'_{Coil}$. All extracted mechanical power is thus converted to the electrical power dissipated by the load and pickup coil, which is an ideal power conversion that can be expressed as

$$P_m = P'_{Coil} + P'_L = 2P'_L. \quad (18)$$

Substituting Eq. (18) into Eq. (1), the output power calculation therefore is corrected by

$$P_L + P_{Coil} = P_m - P_{Iron} \Rightarrow P_L = (2P'_L - P_{Iron})/2, \quad (19)$$

where P_{Iron} denotes the iron loss, which needs to be calculated separately.

Because both magnets in one VREH are stationary, the flux loop can not change its magnetic polarization, which presents a DC-biased flux change in the electrical steel. Knowing the peak-peak flux density change Δ_B , the iron loss can be estimated following the Steinmetz equation [21] with

$$P_{Iron} = \int_{V_{Iron}} k \left(\frac{f_{Coil}}{1\text{Hz}} \right)^\alpha \left(\frac{\Delta_B}{1\text{T}} \right)^\beta dv. \quad (20)$$

Herein, V_{Iron} is the total volume of ferromagnetic material, i.e., the pole-piece and the toothed wheel, and k , α , and β are Steinmetz coefficients depending on the characteristics of the specific material being used. Although information on the exact material utilized in our simulations and experiments (NSC-35H230) is not available, the Steinmetz coefficients of the electrical steel M235-35A is utilized, which is of the same grade [21].

Additionally, the cutting process can negatively affect the magnetic properties of electrical steels, increasing iron losses. For experiments, however, the electrical steel is cut using wire electrical discharge machining (WEDM). It has been shown that this method has the lowest effect on electrical steel, which minimizes iron loss due to fabrication [22,23]. The effects of the cutting process are thus neglected in the estimation of iron loss.

5. Experimental setup

Fig. 7 depicts the experimental setup that is used. In order to minimize unwanted external effects, all mechanical supports and fasteners in proximity to the VREH are made of non-metallic materials, including polyoxymethylene (POM) and acrylic glass. Moreover, the entire setup is placed in an environmental chamber that maintains the ambient temperature at 20 °C.

The shaft of the rotating environment is constructed of a POM cylinder with a diameter of 270 mm, which is driven by a DC motor. In addition, a stainless steel flywheel is mounted behind the cylinder to support stable rotating speeds.

The pole-pieces and toothed wheels are stacked by 28-layers of electrical steel laminations, forming a bulk with a stack length of $l_z = 10.15$ mm. Fig. 8 illustrates exemplar laminations, which are cut from an NSC-35H230 electrical steel sheet by a WEDM.

As previously mentioned, the desired magnet height h_M is implemented using a number of thinner N50-NdFeB magnets [24] with the same width and depth as the pole-piece (see Table 1). The pole-piece and its bulk magnets are clamped in an acrylic housing, exemplified in Fig. 9. This arrangement supports the placement of the pickup unit with the help of an adjustable lift stage, demonstrated in Fig. 10.

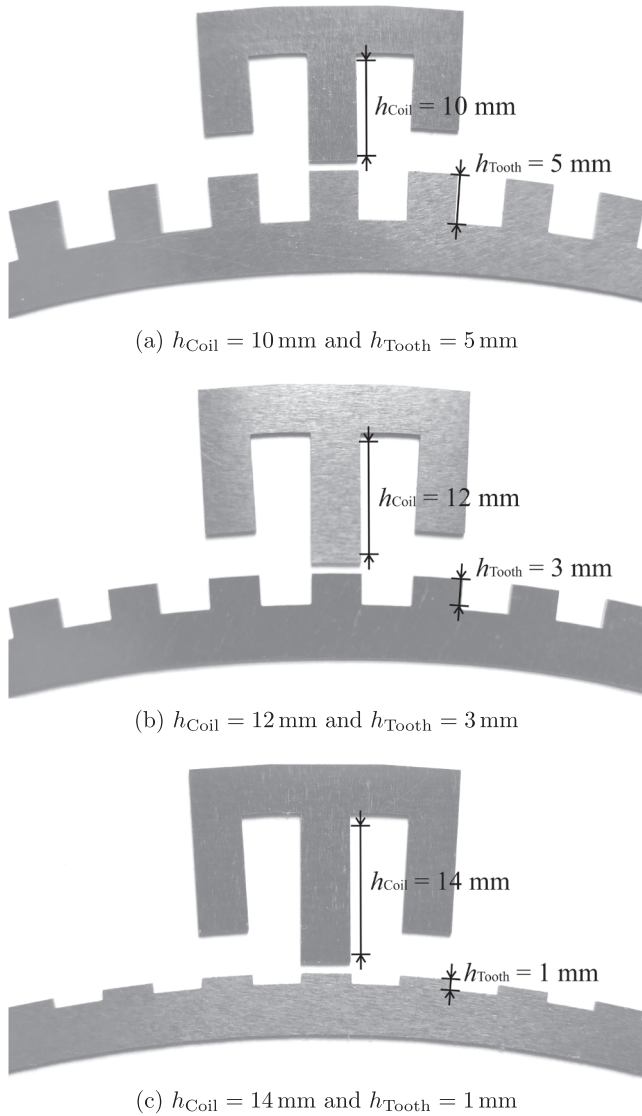


Fig. 8. Exemplar pole-pieces and toothed wheels of three different height combinations. The shown laminations are made of electrical steel (NSC-35H230) and constructed for a magnet height of $h_M = 3.4 \text{ mm}$.

To extract the maximum available power from the VREHs, a complex conjugate matched impedance is required as load to the transducer. A resistor is connected to the pickup coil, and a compensating capacitor is connected in series to the resistor. The required resistance has been obtained by measuring the ohmic-resistance of the copper coil. The capacitance (C_{Tune}), however, is dependant on the pickup coil inductance (L_{Coil}) and electrical frequency of the output voltage, which can be calculated as

$$C_{\text{Tune}} = \frac{1}{(2\pi f_{\text{Coil}})^2 L_{\text{Coil}}} \quad (21)$$

where f_{Coil} is the electrical frequency, which is dominated by the angular velocity and number of teeth ($f_{\text{Coil}} = \omega N_{\text{Tooth}} / (2\pi)$). To obtain the output voltage of the energy harvester, we connect the circuit to an oscilloscope (Keysight MSO-X 3024T). Measurements of the output voltage are conducted over one mechanical rotation cycle, i.e., approximately 6s, which is equivalent to 90 electrical cycles. We then compute the average output power of the obtained electrical cycles, considering the load resistance.

6. Results and discussion

In this section, we present the results of both numerical analysis and experimental verification. In Section 6.1 a comparison between laboratory experiment and numerical simulation is presented. These results demonstrate that the FEA setup can be used for further analysis with a low error. In Section 6.2, we will then utilize the verified FEA setup to investigate the structural optimization of size-constrained VREHs and effects of different angular velocities on the output power. The findings of numerical simulations are supported by individual measurements on the experimental setup. Moreover, in Sec. 6.3, a simple survey is carried out to demonstrate that the VREH can supply sufficient power for low-energy wireless sensors already at low rotational speeds. Finally, the influence of the energy harvester on its rotating host is investigated by analyzing the induced torque on its rotating shaft. Results of this investigation are presented in Sec. 6.4.

6.1. Experimental verification

Fig. 11 shows measurements and simulations of the load voltage for an exemplar VREH structure. In this specific structure, h_{Tooth} is 5 mm, h_{Coil} is 10 mm, and the h_M is 3.4 mm. Moreover, the matched impedance is 60.5Ω .

Fig. 11 demonstrates a good agreement between the simulated and measured output voltage of the harvester, with a calculated MAPE of 2.71%. Based on the connected load, the RMS output power of simulation and experiment can be calculated according to

$$P_{\text{RMS}} = \frac{1}{T} \cdot \frac{\Delta t}{2N} \cdot \sum_{n=0}^N [P(t_n) + P(t_{n+1})] \quad (22)$$

where T is the signal period, Δt is the step size, N is the number of samples, and $P(t_n)$ is the instantaneous power with respect to time. The resulting RMS power values for simulation and experiment are $485.56 \mu\text{W}$ and $480.67 \mu\text{W}$, respectively. The difference between simulation and experiment is approx. $5 \mu\text{W}$, or 1% (see also Table 3, Case 3).

6.2. Structural investigation

Having demonstrated a good agreement between simulation and experiment, the numerical simulation can be utilized to sweep a number of structural parameters.

The first parameter to be investigated is the optimal magnet height h_M . According to Eq. (17), the optimal value for h_M for a given structure can be analytically determined. This, however, requires knowledge of the leakage flux coefficient and the iron loss factor. Instead, a numerical approach is utilized here. Fig. 12 shows the result of an h_M sweep in the FEA. Moreover, the results are supported by six different magnet height cases in experiment. The output of simulation ($P_{L,\text{max}}$ at 3.4 mm) and experiments ($P_{L,\text{max}}$ at 3.5 mm) are in good agreement, and a magnet height of 3.4 mm is used for further investigations.

Fig. 13 shows results of an analysis for h_{Tooth} and h_{Coil} for a given volumetric constraint, defined by $C = 15 \text{ mm}$ ($d_o = 324 \text{ mm}$). Again, an FEA sweep is performed and supported by individual experimental cases (a, b and c). The results demonstrate a maximum output power at a tooth height of approximately 3 mm and a resulting coil height of 12 mm. This result coincides well with the knee-point in Fig. 5.

Table 2 summarizes the output powers of the experimental cases presented in Figs. 12 and 13 for different magnet heights and different tooth-to-coil ratios, respectively. The table, moreover, contains simulation results for the same conditions, demonstrating the FEA's capability to accurately predict output power, with relative errors below 3% in all cases.

Fig. 14 presents the output power for five different space constraints, summarized in Table 3. For each case, the output power for an

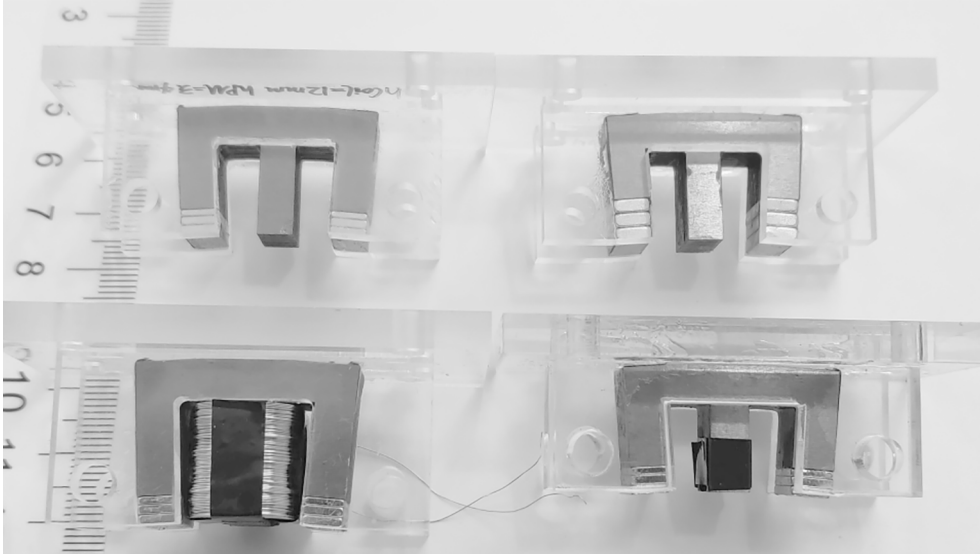


Fig. 9. Pole-piece and magnet bulks are clamped in acrylic housings. A fully assembled pickup unit is shown in the bottom-left.

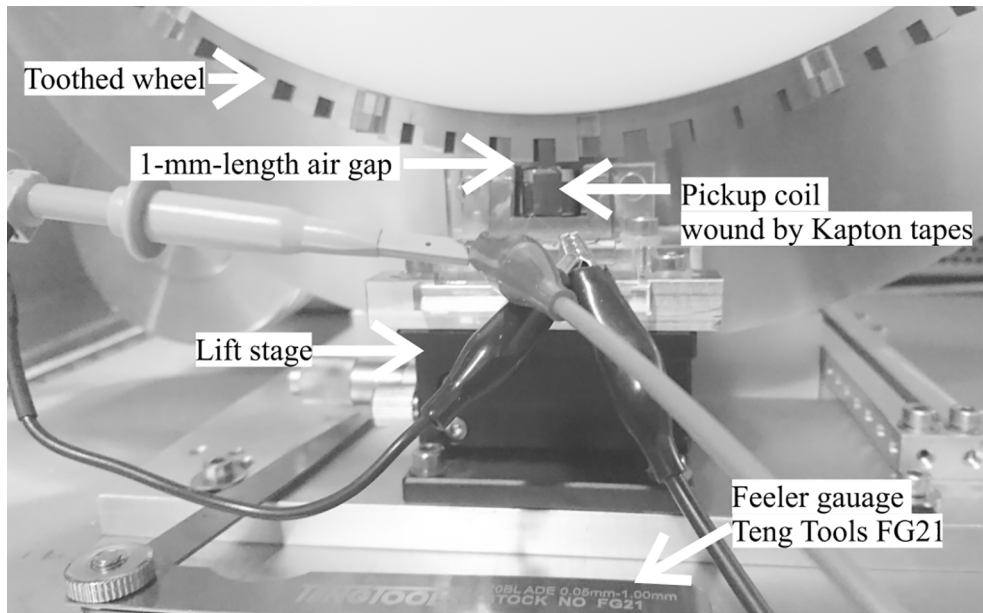


Fig. 10. Close-up of the setup of the pickup unit. The pickup unit is fixed on a lift stage to accurately set the air-gap to 1 mm by using a feeler gauge.

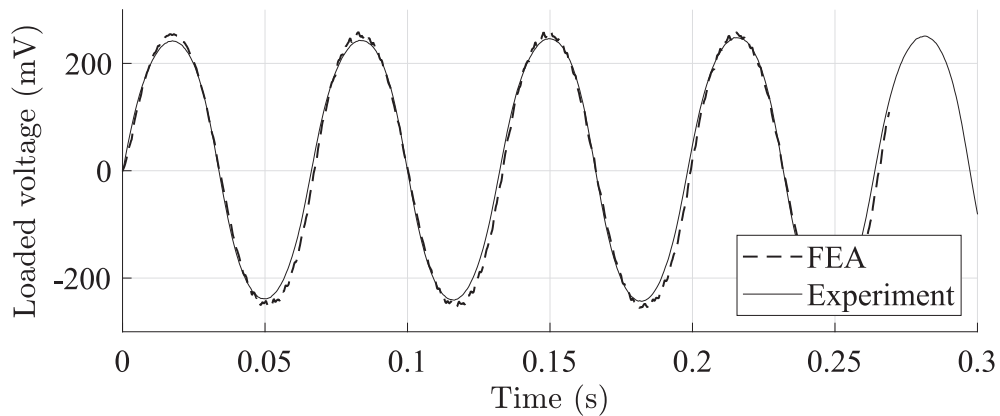


Fig. 11. Comparison of the numerical simulation and experiment results of an exemplar VREH over four electrical cycles. The evaluated structure includes $h_{PM} = 3.4$ mm, $h_{Tooth} = 5$ mm, $h_{Coil} = 10$ mm, and $R_L = 60.5 \Omega$. A good agreement between simulated and measured load voltage was found.

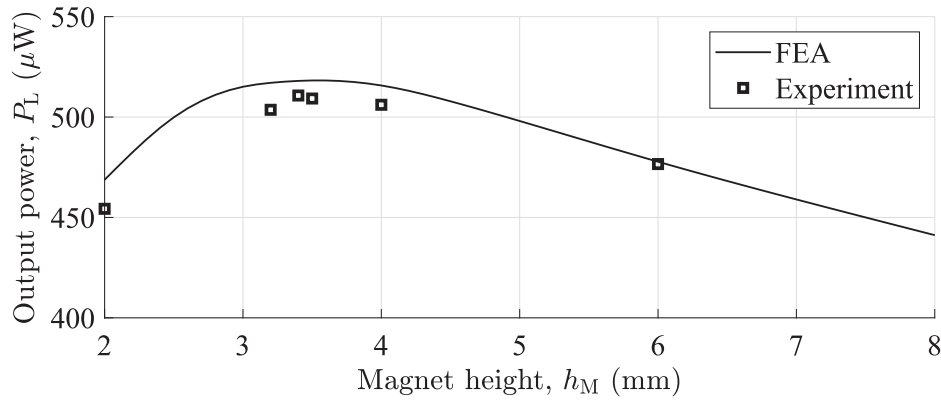


Fig. 12. Output power of a structure with $h_{\text{Coil}} = 12$ mm, $h_{\text{Tooth}} = 3$ mm (i.e. Structure b) based on different magnet heights h_M . Six experimental cases are used to support the FEA results.

optimal utilization of C , that means maximum volume, is presented. Not surprisingly, the output power increases with increasing volume of the harvester. This output power improvement can be normalized with the additional space occupied, resulting in volumetric power densities of $8.45 \mu\text{W cm}^{-3}$ to $11.10 \mu\text{W cm}^{-3}$, according to Eq. (6) and (7). Moreover, with C increasing in a linear manner, even the optimum choices for h_{Coil} and h_{Tooth} follow linear developments. The tooth-to-coil ratio, however, decreases, demonstrating that the contribution of the coil area to the output power is larger. This confirms the expectations, discussed based on the theory in Section 3, that the tooth height only needs to obtain a certain critical value, after which its increment only has minor influences. In contrast, the coil height continues to increase output power as the active coil volume increases.

Finally, the energy harvesters dependency on rotation speed has been evaluated for low rates of rotation. Fig. 15 shows the output power of an exemplar case (Case 3 in Table 3) as a function of the angular velocity. The pickup coil poses a large inductive component ($L_{\text{Coil}} \approx 210 \text{ mH}$), and increasing angular velocities result in higher electrical frequencies of the alternating output signal. In order to utilize the transducer in the best manner, a complex conjugate impedance matching between the pickup coil and load is necessary to achieve maximum power transfer. However, at low frequencies the load can be simplified to be purely resistive, neglecting a frequency dependency.

The result shows that the complex conjugate matching has neglectable influences below 2.1 rad s^{-1} ($\approx 20 \text{ rpm}$). Above this angular frequency, however, a purely resistive load prohibits full exploitation of the harvester's capability, requiring instead a full impedance matching for maximum power transfer. In the evaluated range of angular velocities (equivalent to approx. 5 rpm–60 rpm), the estimated output power reaches up to 22 mW at complex conjugate matching.

Table 2

Power output comparison between simulation and experiment.

Structures		P_L (μW)		Error (%) [*]
		FEA	Exp.	
Structures	a	485.46	480.67	0.99
	b	517.57	511.72	1.14
	c	330.32	323.13	2.23
Magnet Heights h_M [mm]	2.0	468.76	455.30	2.96
	3.2	517.04	504.60	2.47
	3.4	517.57	511.72	1.14
	3.5	518.22	510.22	1.16
	4.0	515.72	507.11	1.70
	6.0	477.75	477.59	0.03

^{*} With respect to the experimental results (Exp.).

6.3. Low-energy wireless sensor requirements

The proposed energy harvesting method aims to supply power to low-energy electronic systems, such as wireless sensors. It is therefore important to evaluate whether the optimized VREH can provide sufficient power levels for duty-cycled, low-energy wireless sensor systems. However, not all of the harvested energy can be made available to the intended load. Instead, a portion is inevitably consumed by conversion circuits, necessary to regulate the output of the transducer to be compatible with the load system. Moreover, this additional circuitry can result in impedance mismatches, ultimately lowering energy extraction.

As the exact implementation of the electrical conversion subsystem, and thus the losses it entails, are application-dependent, an overall extraction efficiency is a common generalization. The power extraction

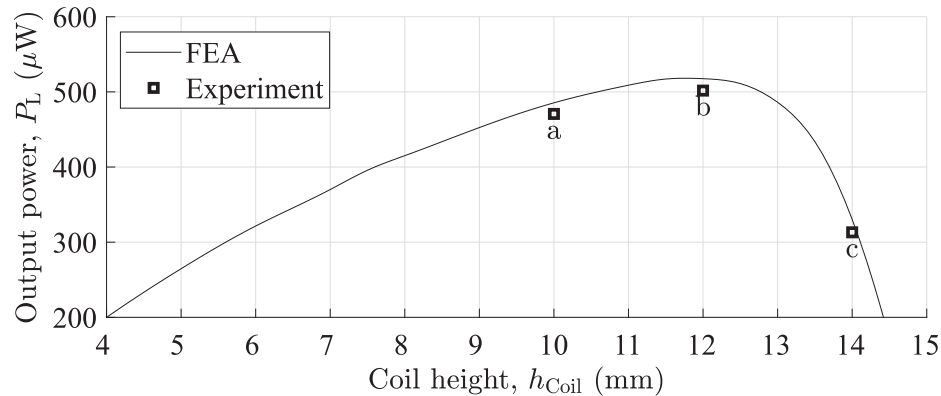


Fig. 13. Output power of VREHs with the same overall diameter $d_o = 324$ mm and magnet height $h_M = 3.4$ mm as a function of different h_{Coil} and h_{Tooth} . Three experimental cases are used to support the FEA results.

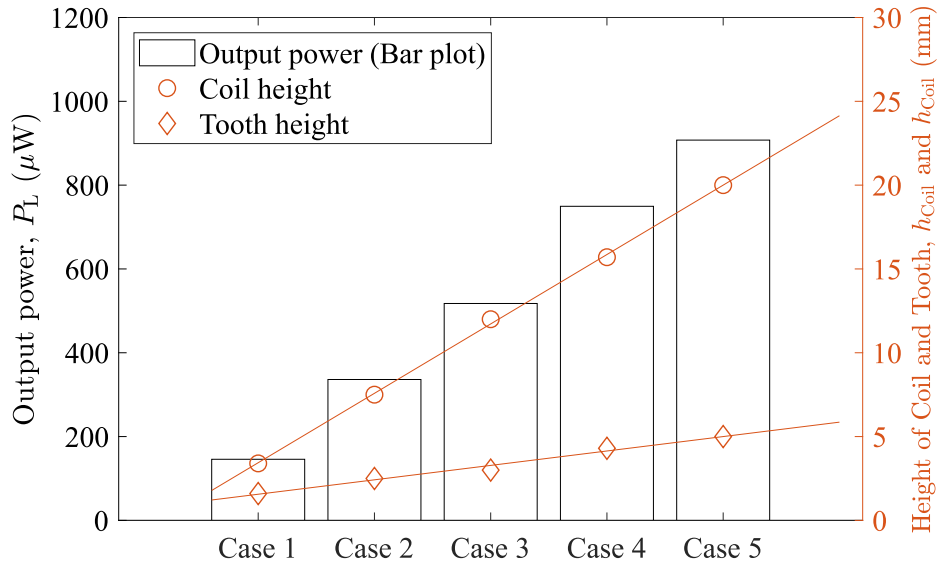


Fig. 14. Simulated output power for five optimized size constraints. All cases are evaluated for a constant angular velocity $\omega = 1.06 \text{ rad s}^{-1}$.

Table 3

The optimal structure of five cases with different overall diameters.

Case	d_o (mm)	C^* (mm)	h_{Coil} (mm)	h_{Tooth} (mm)	h_M (mm)
1	304	5	3.4	1.6	3.4
2	314	10	7.5	2.5	3.4
3	324	15	12.0	3.0	3.4
4	334	20	15.7	4.3	3.5
5	344	25	20.0	5.0	3.5

* With $C = h_{\text{Coil}} + h_{\text{Tooth}}$.

efficiency (PEE) [25] can be used in this regard. In order to provide sufficient power to the load, the following condition needs to be met [26]:

$$P_L \cdot \text{PEE} \geq P_{\text{Req}}. \quad (23)$$

Herein, P_L is the output power of the transducer, and P_{Req} is the power required by the wireless sensor.

Although PEE values are difficult to generalize due to their dependence on harvester, circuit and load implementation, typical values in

the range of 70%–90% have been reported [27]. In this paper, we use a pessimistic value of 60% to investigate the suitability of the proposed VREH structure for five wireless sensor examples, previously reported in the literature. Table 4 lists the requirements of these five cases and the minimum rotational speed for the optimized VREH (Case 3) necessary to match this requirement. The results of this analysis show that sufficient power can be obtained by the VREH at rotational speeds in the range of about 6 rpm–25 rpm.

6.4. Induced torque by the VREH

As a rotational energy harvester, such as the VREH proposed in this work, converts part of the rotational energy of its host into electrical energy for a low-energy electronic system, it can be interpreted as a parasitic system. A mechanical impact on the rotating shaft is the torque ripple caused by the rotational interactions between the pickup unit and the toothed wheel mounted on the shaft. The torque ripple can lead to mechanical vibrations and acoustic noise, which is mostly undesired in rotating shafts. Its contribution can be separated into two

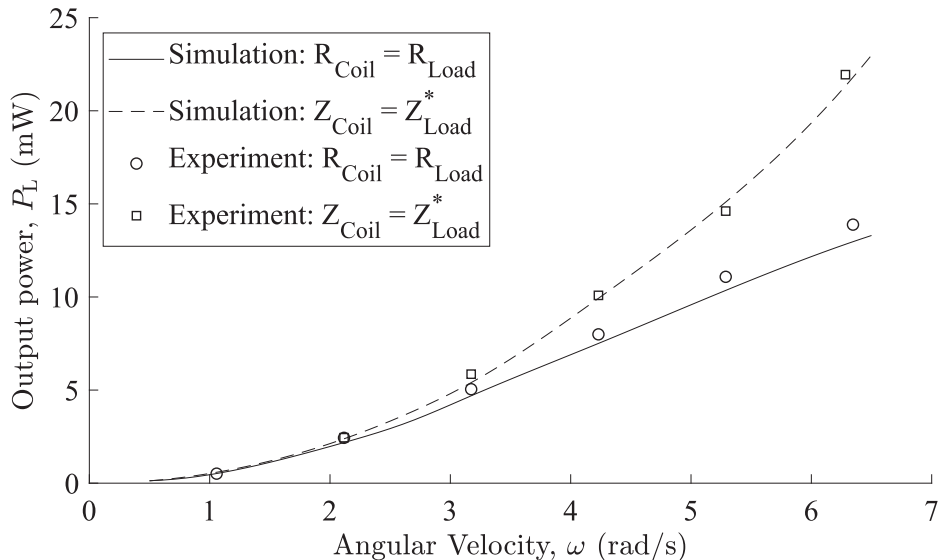


Fig. 15. Simulated output power of the optimized structure of Case 3 as a function of the angular velocity. The experiment results verify the output power at different rotating speeds under the resistive matching ($R_{\text{Coil}} = R_{\text{Load}}$) and complex conjugate matching ($Z_{\text{Coil}} = Z_{\text{Load}}^*$), respectively.

Table 4
Duty-cycled, low-energy wireless sensor systems powered by VREH

Refs.	Application	Sample interval (s)	Power consumption (μ W)	Rotational speed* (rpm)
[28]	Temperature	1	80.91	5.7
[29]	Strain sensing	60	274	9
[30]	Light, temperature, humidity	300	320	11
[29]	Vibration	15	1730	22.5
[31]	Acoustic emission	60	1760	23

* Minimal rotational speed of VREH required by wireless sensors.

components, namely the electromagnetic torque and the cogging torque.

The former has been introduced in Section 2.2, a torque caused by the electromagnetic force generated by the induced current in the pickup coil, which contributes to the conversion of the kinetic energy into electrical energy. Cogging torque, instead, is caused by the magnets in the pickup unit and is primarily dependent on their volume and geometry, as well as the pole-piece and toothed wheel [10].

In order to investigate the torque ripple of the VREHs in this work, Arkkio's method is used in COMSOL [10]. Fig. 16–18 present the results of this investigation, depicting the effects of different VREH geometrical properties.

Fig. 16 shows that, while using the same magnet size (i.e., $h_M = 3.4$ mm), the toothed wheel geometry, specifically the tooth height, dominates the torque ripple. The structures (a) and (b) with longer teeth result in a higher magnitude of the torque ripple (≈ 0.7 M·m) than that of structure (c). However, similarly to the output power, the effect of increasing tooth height appears to saturate even on the induced torque. As a result, structure (a) and (b) show a nearly identical torque ripple, despite their differences in tooth heights.

In Fig. 17, influences of the magnet height is demonstrated for the optimized structure of Case 3 (i.e., $h_{Coil} = 12$ mm and $h_{Tooth} = 3$ mm). The results show an increasing torque amplitude with increasing magnet height. Increasing the magnet height from 2 mm to 6 mm doubles the torque amplitude.

Fig. 18 depicts the torque ripples caused by three optimized structures with different size constraints (see Table 3). The three structures, especially Case 3 and Case 5, result in a torque ripple showing no major differences, because they are using magnets with similar geometrical size (i.e., $h_M \approx 3.5$ mm). The slightly lower torque ripple caused by Case 1 is a result of its shorter teeth ($h_{Tooth} = 1.6$ mm), which reduces the

variable reluctance effect.

7. Conclusion

In this paper, a novel VREH topology based on an m-shaped pole-piece and two opposing magnets is investigated in detail. The goal with this topology is to provide an energy harvester for low-speed rotational kinetic motion that can supply power to wireless sensor systems. Moreover, we provide a structural optimization to design a high-volumetric-power-density VREH, which allows to achieve a compact structure for embedded or size-constrained applications. A theoretical analysis based on the VREH operation principle is presented, which shows that the output power depends mainly on three geometrical factors, namely the coil height, the tooth height, and the magnet height. Moreover, some of these factors require trade-offs if the application scenario poses a size-constraint. In order to evaluate influences on the output power, and to optimize the geometric parameter selection for a maximum power density, we performed a numerical analysis, verified by experimental cases.

With less than 3% error in power estimations, the results demonstrate a good agreement between numerical simulation and experiments in all cases that have been evaluated. This provides trust in the implemented FEA model and the simulation setup, which allows to investigate a number of structural and application configurations in detail.

We were able to show that for a given structure and size constraint, optimal values for the magnet height, the tooth height, and the coil height could be found. The determined values define the implementation of the proposed VREH structure in order to maximize its power output under given constraints. This work, therefore, provides a numerical method for the structural optimization of the m-shaped VREH. The importance of such an optimization is confirmed by the presented experimental results, demonstrating a high sensitivity to geometrical dimensions, which risks output power reductions if not properly selected.

We, moreover, present how the output power scales for different size constraints and different angular velocities, which demonstrates that the structure is scalable to different application scenarios. The proposed and optimized VREH structure generates output powers of 0.12 mW–22 mW at 5 rpm–60 rpm and a size constraint of 43.5 cm³. This is sufficient power for a number of industrial sensing tasks, and thus enables self-powered condition monitoring solutions in rotating applications without the need of battery replacements.

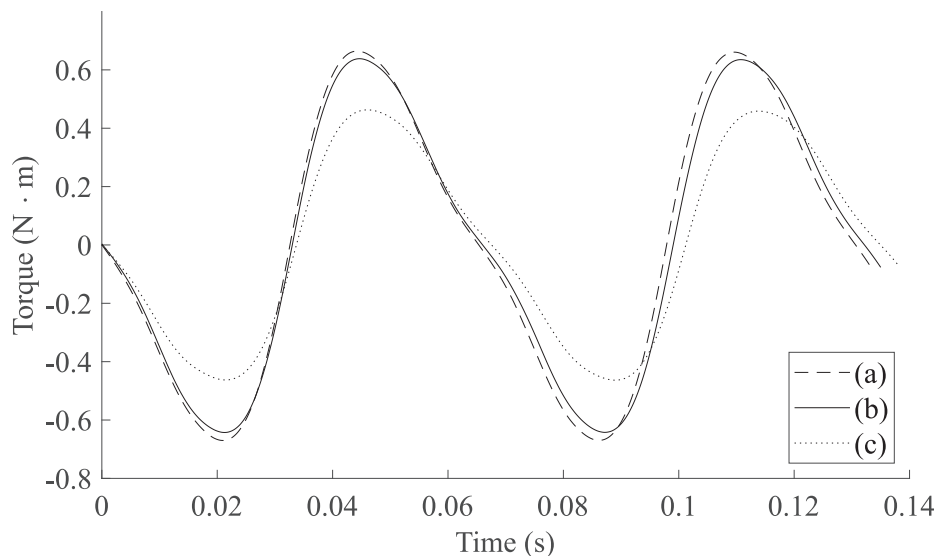


Fig. 16. Torque ripple caused by the three structures with different tooth and coil heights but the same magnet height (i.e., (a)–(c) of Case 3 shown in Fig. 8).

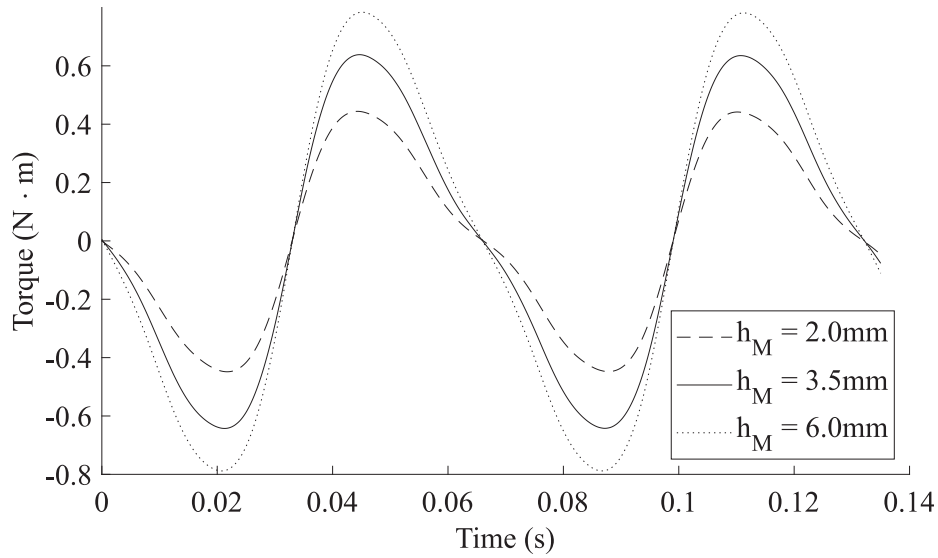


Fig. 17. Torque ripple caused by the optimized structure Case 3 (i.e., $h_{\text{Coil}} = 12$ mm and $h_{\text{Tooth}} = 3$ mm) with three different magnet heights.

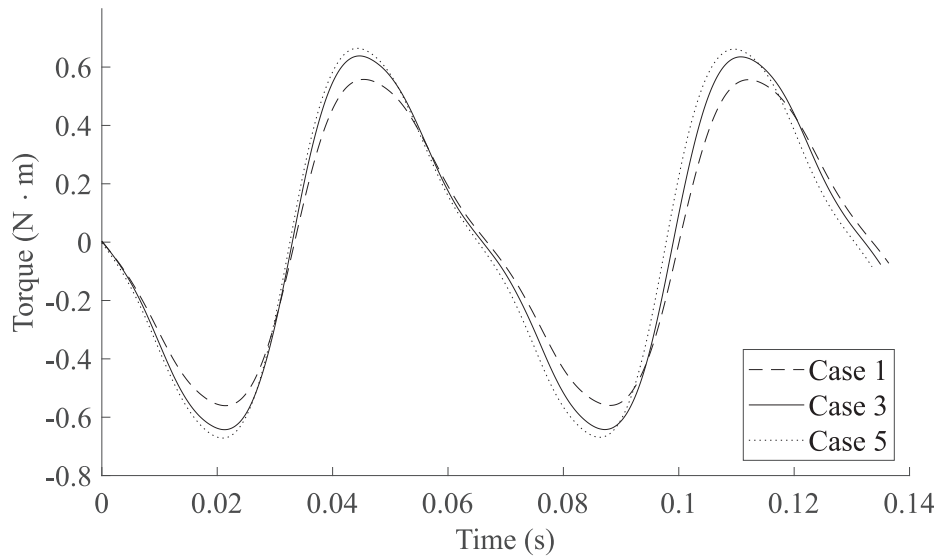


Fig. 18. Torque ripple caused by three different optimized structures with a similar magnet height (i.e., Case 1, Case 3, and Case 5 shown in Table 3).

Finally, we investigated the torque ripple, induced by the VREH on its rotating host. This investigation demonstrated that the torque amplitude is dominated by contributions of cogging torque, which in itself is mainly defined by magnet geometries. Together with the VREH optimization, this study thus provides a guideline for users to design an appropriate structure for their applications by considering size constraints, output power, and induced torque.

Declaration of Competing Interest

None declared.

Acknowledgements

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Appendix A. Derivation of coil influence on output power

The coil resistance R_{Coil} in Eq. 3a is obtained by

$$R_{\text{Coil}} = \frac{4\rho_{\text{wire}}N_{\text{Coil}}\overline{C_{\text{turn}}}}{\pi d_{\text{wire}}^2}, \quad (\text{A.1})$$

where ρ_{wire} is the electrical resistivity of the wire material, d_{wire} is the bare wire diameter, and $\overline{C_{\text{turn}}}$ is the mean length over all turns N_{Coil} expressed by Eq. (A.2).

$$\overline{C_{\text{turn}}} = \pi(r_o + r_i) + 2(l_z + W_{\text{cp}}), \quad (\text{A.2})$$

where r_o and r_i are the outer and inner radii of the coil's end part shown in Fig. 2a, and W_{cp} is the center pole width. Substituting Eq. (A.1) into Eq. (3a), yields

$$P_L \propto \frac{\pi N_{Coil} d_{wire}^2}{4 \rho_{wire} C_{turn}} \quad (A.3)$$

The product $N_{Coil} d_{wire}^2$ in Eq. (A.3) is given by:

$$N_{Coil} d_{wire}^2 = \frac{4 \mathcal{V}_{Coil}}{\pi D_{xz}}, \quad (A.4)$$

where $\mathcal{V}_{Coil}$ and D_{xz} are the coil volume and the coil dimension factor expressed by Eqs. (A.5) and (A.6), respectively.

$$\mathcal{V}_{Coil} = h_{Coil} [\pi (r_o^2 - r_i^2) + 2 W_{Coil} (l_z + W_{cp})], \quad (A.5)$$

where W_{Coil} and h_{Coil} are the coil width and height, respectively, shown in Fig. 2c, and an ideal rectangular coil has $W_{Coil} = r_o - r_i$.

$$D_{xz} = \pi (r_o + r_i) + (l_z + W_{cp}), \quad (A.6)$$

where D_{xz} is related to the coil dimensions of xz-plane shown in Fig. 2a (i.e., r_o , r_i , W_{cp} and l_z).

Combining Eqs. A.2, A.5, A.4, A.3 and assuming D_{xz} is a fixed value, P_L in Eq. (3a) can be simplified to

$$P_L \propto \frac{h_{Coil} W_{Coil}}{\rho_{wire}}. \quad (A.7)$$

In a practical coil winding, the insulation of the magnet wire causes the area $h_{Coil} W_{Coil}$ to not fully be occupied by copper. Assuming ρ_{wire} is a constant, Eq. (A.7) thus is rewritten as

$$P_L \propto h_{Coil} W_{Coil} f_w : f_w < 1, \quad (A.8)$$

where f_w is the winding fill factor. Assuming a round magnet wire to wind the pickup coil, f_w is defined as [32]

$$f_w = S_1 S_2 S_i : S_1 = \left(\frac{d_{wire}}{d_e} \right)^2, \quad S_2 \approx 0.91, \quad S_c < 1, \quad \text{and} \quad S_i < 1, \quad (A.9)$$

where S_1 is the squared ratio of the bare wire diameter d_{wire} to the overall wire diameter d_e , and S_2 is the multi-layer coil winding fill factor, which ideally is achieved by an orthocyclically wound coil. Since forming such an ideal one is not available, adding S_i to Eq. (A.9) adjusts for a real coil winding fill factor. S_i is the ratio of the coil area (i.e., the product of $h_{Coil} W_{Coil}$) to the slot area of the pickup-unit. It is necessary to apply an electrical insulation (i.e., a coil bobbin) between the pickup coil and the pole-piece, and S_i can indicate how much of the available area is actually being used for the insulation.

Appendix B. Derivation of optimized magnet height

According to Eq. (15), $\mathcal{R}_M$, $\mathcal{R}_{g,a}$, and $\mathcal{R}_{g,u}$ in Eq. (16) are expressed as

$$\mathcal{R}_M = \frac{h_M}{\mu_0 \mu_M A_M}, \quad \text{and} \quad A_M = l_z W_M \quad (B.1)$$

$$\mathcal{R}_{g,x} = \frac{l_{g,x}}{\mu_0 \mu_{g,x} A_{g,x}}, \quad \text{and} \quad A_{g,x} = l_z W_{g,x}, \quad (B.2)$$

where x denotes either a or u for the aligned and unaligned position, respectively.

Assuming a uniform flux distribution in the magnet, because of the rectangular block shape, the magnet reluctance can be derived by Eq. (B.1).

Due to the fringing effect, the air-gap reluctance can be divided into n separate elements $\mathcal{R}_i$, which provide the total reluctance according to $\mathcal{R}_{g,x} = \sum_{i=1}^n \mathcal{R}_{i,x}$ [16]. Applying Carter's theory [33], $\mathcal{R}_{g,x}$ expressed by Eq. (B.2) is assumed as a normalized reluctance component having a uniform flux distribution in the air gap. Assuming l_z is a fixed value, the dimensions $l_{g,x}$ and $W_{g,x}$ thus are equivalent length and width of the air-gap reluctance, respectively.

Moreover, $\mathcal{F}_M$ is given by Eqs. (5) and (16) yields a function with respect to h_M as

$$\Delta_\phi(h_M) = \mu_0 h_M H_C \frac{\left(\frac{l_{g,u}}{A_{g,u}} - \frac{l_{g,a}}{A_{g,a}} \right)}{\left(\frac{h_M}{\mu_M A_M} + \frac{l_{g,a}}{A_{g,a}} \right) \left(\frac{h_M}{\mu_M A_M} + \frac{l_{g,u}}{A_{g,u}} \right)}. \quad (B.3)$$

Let the derivative of Eq. (B.3) be equal to zero, the optimal h_M that maximizes Δ_ϕ can be found according to

$$\frac{d\Delta_\phi(h_M)}{dh_M} = 0 \Rightarrow h_M = \mu_M A_M \sqrt{\frac{l_{g,a} l_{g,u}}{A_{g,a} A_{g,u}}}, \quad (B.4)$$

where A is given by Eqs. (B.1) and (B.2). $W_{g,a}$ is equivalent to W_{Path} at the aligned position, and according to Eq. (8), $W_{g,a}$ thus is $W_{g,a} = W_M$. Consequently, Eq. (B.4) can be simplified to

$$h_M = \mu_M \sqrt{\frac{W_{g,a}}{W_{g,u}}} l_{g,a} l_{g,u} \quad (B.5)$$

Because of the finite permeability of the soft iron, the leakage flux and the MMF drop in the soft iron are unavoidable in the VREH, which effects

the magnet's operating condition [34]. Two correction factors should be considered for correcting the h_M estimation, and Eq. (B.5) thus yields

$$h_M = \mu_M \frac{k_2}{k_1} \sqrt{\frac{W_{g,a}}{W_{g,u}}} l_{g,a} l_{g,u}, \quad (\text{B.6})$$

where k_1 is the flux leakage coefficient, k_2 is the loss factor, and both factors are bigger than 1.

Appendix C. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.enconman.2019.05.082>.

References

- [1] Babayo AA, Anisi MH, Ali I. A review on energy management schemes in energy harvesting wireless sensor networks. *Renew Sustain Energy Rev* 2017;76:1176–84. <https://doi.org/10.1016/J.RSER.2017.03.124>.
- [2] Sherazi HHR, Grieco LA, Boggia G. A comprehensive review on energy harvesting MAC protocols in WSNs: challenges and tradeoffs. *Ad Hoc Networks* 2018;71:117–34. <https://doi.org/10.1016/J.ADHOC.2018.01.004>.
- [3] Beeby S, White N. *Energy Harvesting for Autonomous Systems*. Artech House 2010:45–155. (Ch. 3–5).
- [4] Shaikh FK, Zeadally S. Energy harvesting in wireless sensor networks: a comprehensive review. *Renew Sustain Energy Rev* 2016;55:1041–54. <https://doi.org/10.1016/J.RSER.2015.11.010>.
- [5] Guan M, Liao W-H. Design and analysis of a piezoelectric energy harvester for rotational motion system. *Energy Convers Manage* 2016;111:239–44. <https://doi.org/10.1016/j.enconman.2015.12.061>.
- [6] Zou H-X, Zhang W-M, Li W-B, Wei K-X, Gao Q-H, Peng Z-K, Meng G. Design and experimental investigation of a magnetically coupled vibration energy harvester using two inverted piezoelectric cantilever beams for rotational motion. *Energy Convers Manage* 2017;148:1391–8. <https://doi.org/10.1016/j.enconman.2017.07.005>.
- [7] Febbo M, Machado S, Gatti C, Ramirez J. An out-of-plane rotational energy harvesting system for low frequency environments. *Energy Convers Manage* 2017;152:166–75. <https://doi.org/10.1016/j.enconman.2017.09.042>.
- [8] Siddique ARM, Mahmud S, Heyst BV. A comprehensive review on vibration based micro power generators using electromagnetic and piezoelectric transducer mechanisms. *Energy Convers Manage* 2015;106:728–47. <https://doi.org/10.1016/j.enconman.2015.09.071>.
- [9] Zhang Y, Cao J, Zhu H, Lei Y. Design, modeling and experimental verification of circular halbach electromagnetic energy harvesting from bearing motion. *Energy Convers Manage* 2019;180:811–21. <https://doi.org/10.1016/j.enconman.2018.11.037>.
- [10] Xu Y, Bader S, Oelmann B. A survey on variable reluctance energy harvesters in low-speed rotating applications. *IEEE Sens J* 2018;18(8):3426–35. <https://doi.org/10.1109/JSEN.2018.2808377>.
- [11] Parthasarathy D, Enoksson P, Johansson R. Prototype energy harvesting wheel speed sensor for anti-lock braking. *Proceedings of IEEE International Symposium on Robotic and Sensors Environments* 2012. p. 115–20. <https://doi.org/10.1109/ROSE.2012.6402607>.
- [12] Kroener M, Ravindran SKT, Woias P. Variable reluctance harvester for applications in railroad monitoring. *J Phys: Conf Ser* 2013;476(1):012091. <https://doi.org/10.1088/1742-6596/476/1/012091>.
- [13] Kroener M, Moll N, Ravindran SKT, Mehne P, Woias P. Characterization of a variable reluctance harvester. *J Phys: Conf Ser* 2014;557(1):012035. <https://doi.org/10.1088/1742-6596/557/1/012035>.
- [14] Moss SD, Payne OR, Hart GA, Ung C. Scaling and power density metrics of electromagnetic vibration energy harvesting devices. *Smart Mater Struct* 2015;24(2):023001. <https://doi.org/10.1088/0964-1726/24/2/023001>.
- [15] Balakrishnan A, Joines WT, Wilson TG. Air-gap reluctance and inductance calculations for magnetic circuits using a schwarz-christoffel transformation. *IEEE Trans Power Electron* 1997;12(4):654–63. <https://doi.org/10.1109/63.602560>.
- [16] Lazarov V, Rizov P. Magnetic permeance of the air gap between two toothed structures. *J Mater Process Technol* 2005;161(1):96–100. <https://doi.org/10.1016/j.jmatprotec.2004.07.059>.
- [17] Mao S, Dorrell D, Tsai M. Fast analytical determination of aligned and unaligned flux linkage in switched reluctance motors based on a magnetic circuit model. *IEEE Trans Magn* 2009;45(7):2935–42. <https://doi.org/10.1109/TMAG.2009.2016087>.
- [18] Skomski R, Coey J. Magnetic anisotropy – how much is enough for a permanent magnet? *Scr Mater* 2016;112:3–8. <https://doi.org/10.1016/j.scriptamat.2015.09.021>.
- [19] Demenko A, Wojciechowski RM, Sykulski JK. 2-D versus 3-D electromagnetic field modeling in electromechanical energy converters. *IEEE Trans Magn* 2014;50(2):897–900. <https://doi.org/10.1109/TMAG.2013.2282415>.
- [20] Nippon Steel and Sumitomo Metal. Non-oriented electrical steel sheets [Online; accessed Feb-2019]; 2015.
- [21] Flankl M, Tüysüz A, Kolar JW. Cogging torque shape optimization of an integrated generator for electromechanical energy harvesting. *IEEE Trans Indus Electron* 2017;64(12):9806–14. <https://doi.org/10.1109/TIE.2017.2733441>.
- [22] Kurosaki Y, Mogi H, Fujii H, Kubota T, Shiozaki M. Importance of punching and workability in non-oriented electrical steel sheets. *J Magn Magn Mater* 2008;320(20):2474–80. <https://doi.org/10.1016/J.JMMM.2008.04.073>.
- [23] Hofmann M, Naumoski H, Herr U, Herzog HG. Magnetic properties of electrical steel sheets in respect of cutting: micromagnetic analysis and macromagnetic modeling. *IEEE Trans Magn* 2016;52(2):1–14. <https://doi.org/10.1109/TMAG.2015.2484280>.
- [24] Webcraft GmbH. N50 NdFeB Magnet [Online; accessed Feb-2019]; 2011.
- [25] Ngo KD, Phipps A, Nishida T, Lin J, Xu S. Power converters for piezoelectric energy extraction. *ASME 2006 international mechanical engineering congress and exposition American Society of Mechanical Engineers*; 2006. p. 597–602. <https://doi.org/10.1115/IMECE2006-14343>.
- [26] Martinez B, Montn M, Vilajosana I, Vilajosana X. Early scavenger dimensioning in wireless industrial monitoring applications. *IEEE Internet Things J* 2016;3(2):170–8. <https://doi.org/10.1109/JIOT.2015.2483750>.
- [27] Szarka GD, Stark BH, Burrow SG. Review of power conditioning for kinetic energy harvesting systems. *IEEE Trans Power Electron* 2012;27(2):803–15. <https://doi.org/10.1109/TPEL.2011.2161675>.
- [28] Toh WY, Tan YK, Koh WS, Siek L. Autonomous wearable sensor nodes with flexible energy harvesting. *IEEE Sens J* 2014;14(7):2299–306. <https://doi.org/10.1109/JSEN.2014.2309900>.
- [29] Torfs T, Sterken T, Brebels S, Santana J, van den Hoven R, Spiering V, Bertsch N, Trapani D, Zonta D. Low power wireless sensor network for building monitoring. *IEEE Sens J* 2013;13(3):909–15. <https://doi.org/10.1109/JSEN.2012.2218680>.
- [30] Wang W, Wang N, Jafer E, Hayes M, O'Flynn B, O'Mathuna C. Autonomous wireless sensor network based Building Energy and Environment Monitoring system design. 2010 the 2nd conference on environmental science and information application technology 2010. p. 367–72. <https://doi.org/10.1109/ESIAT.2010.5568311>. vol. 3.
- [31] Delgado Prieto M, Zurita Millan D, Wang W, Machado Ortiz A, Ortega Redondo JA, Romeral Martinez L. Self-powered wireless sensor applied to gear diagnosis based on acoustic emission. *IEEE Trans Instrum Meas* 2016;65(1):15–24. <https://doi.org/10.1109/TIM.2015.2476278>.
- [32] McLyman C. *Transformer and inductor design handbook*. 4th ed. CRC Press; 2016. Ch. 4. pp. 4.1–4.19.
- [33] Calixto WP, Alvarenga B, Coimbra AP, Alves AJ, Martins Neto L, Wu M, da Silva WG, Delbone E. Carter's factor calculation using domain transformations and the finite element method. *Int J Numer Modell: Electron Netw Devices Fields* 2012;25(3):236–47. <https://doi.org/10.1002/jnm.827>.
- [34] Campbell P. *Permanent magnet materials and their application*. Ch. 4. Cambridge University Press; 1996. p. 82–97.